# Ref. [30] — Self-Stabilization at the Edge of Stability

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## 1. Problem: why does large-step gradient descent not simply diverge?

Cohen et al. observed *progressive sharpening*: the top Hessian eigenvalue rises until its product with the learning rate is around two, then often hovers near this boundary while training loss keeps falling non-monotonically. Damian et al. seek an analytical mechanism for this surprising behavior, not merely another measurement.

## 2. Derive the stability threshold

Near a stationary point $\theta^*$, write $e_t = \theta_t - \theta^*$ and approximate $\nabla L(\theta_t) \simeq H e_t$. GD gives

$$e_{t+1} \simeq (I - \eta H) e_t.$$

Along a positive-curvature eigenvector with eigenvalue $\lambda$, the component multiplies by $1 - \eta\lambda$. Local stability needs $|1 - \eta\lambda| < 1$, or $0 < \eta\lambda < 2$. Define *sharpness* $S(\theta) = \lambda_{\max}(\nabla^2 L(\theta))$; the relevant top-direction boundary is $S \simeq 2/\eta$.

## 3. The third-order mechanism

Let $u$ denote a locally selected top eigenvector and $x$ the signed displacement in direction $u$. A local expansion in that direction includes terms beyond quadratic:

$$L(\theta + xu) \approx L(\theta) + x\,\nabla L(\theta)^T u + \frac{x^2}{2} S(\theta) + O(x^3).$$

Differentiating with respect to the slow coordinates produces a curvature-gradient coupling of approximate size

$$\nabla_\theta L(\theta + xu) \supset \frac{x^2}{2} \nabla S(\theta), \quad \Delta\theta_{\text{slow}} \supset -\eta \frac{x^2}{2} \nabla S(\theta).$$

When $S > 2/\eta$, oscillations along $u$ grow; the quadratic-in-amplitude drift pushes toward lower sharpness. Oscillations can then contract when stability is restored. This is the self-stabilizing feedback: the instability itself helps change the landscape location.

**Main result.** With suitable analytical conditions, the authors show that GD near the edge of stability is closely approximated by dynamics resembling projected gradient descent constrained by $S(\theta) \le 2/\eta$. They make quantitative predictions for loss, sharpness and deviations, tested theoretically in their regime and empirically on training examples.

## 4. The important distinction: observed vs proved

The article's experiments motivate a broad phenomenon, but the proof uses a controlled local description, smoothness and other assumptions. "GD tracks constrained descent" is an *approximate dynamical explanation*, not a universal theorem that any architecture, optimizer, stochastic minibatch or step size converges globally.

**Survey relevance (RQ2, RQ4).** Classical small-step arguments require descending loss each step. Here GD is often non-monotone; the coupled sharpness–oscillation dynamics explain why the longer-time loss can still decrease. Do not conflate the stability set $\{S \le 2/\eta\}$ with a valley floor or a set of globally optimal parameters.

## 5. Our notes

The new idea is that going unstable in a sharp direction can automatically push GD into a less sharp part of the loss surface. That offers a mechanism for the edge of stability, but it depends on a particular local smooth model and should not be quoted as a blanket guarantee for Adam or all large-step training. **Original source:** https://openreview.net/pdf/5ce767b4f951bb7db5b2c04af424d1c59c1e9387.pdf.